AeroNest OK-90 Oxygen Therapy Kennel

Domstad Veterinary Teaching Clinic, Utrecht · Clinical Engineering Section · DS-ICU-017 rev. C · Issued 14 September 2026 · Replaces rev. B of 2 February 2025

1. IDENTIFICATION AND SCOPE

This data sheet describes the two AeroNest OK-90 oxygen therapy kennels installed in the small-animal intensive care ward (room B1.07, ground floor of the Hoefslag building) of the Domstad Veterinary Teaching Clinic. The kennels are made by Zeldenrust Medisch B.V., Amersfoort, and are registered under clinic asset numbers `DVTC-ICU-0412` (left bay) and `DVTC-ICU-0413` (right bay). The smaller OK-60 cat kennel in the isolation unit (room B2.19) is covered by sheet DS-ICU-018. Rev. C adds the CO_2 flow rule (section 4) and the brachycephalic limits (section 6). Calculations are planning aids only; the built-in oxygen analyser reading always takes precedence.

2. TECHNICAL SPECIFICATIONS

- **Internal dimensions:** 90 cm wide × 65 cm deep × 70 cm high (gross volume 409.5 L). **Net gas volume** with the floor mattress, drip tray and cooling duct fitted: **V = 395 L**.
- **Patient size:** dogs and cats up to 40 kg body mass; maximum floor load 60 kg. Larger dogs receive nasal oxygen (ICU protocol P-14).
- **Door:** polycarbonate front door with two 12 cm iris ports for handling without opening the door.
- **Gas inlets:** oxygen inlet with NIST connector, supplied at 4.0 bar from the wall outlet; flow meter range 0 to 30 L/min; second inlet for medical air; nebuliser port `NB-2`.
- **Oxygen analyser:** galvanic cell, range 0 to 100 % O_2, accuracy ±2 % vol; sensor cartridge `ZM-OX-31`, replace every 18 months or when the 100 % calibration fails.
- **Climate module:** Peltier cooling, 350 W, chamber temperature adjustable from 16 °C to 28 °C; humidity alarm above 70 % RH.
- **CO_2 absorber:** rear canister holding 1.2 kg soda lime (granules 2 to 4 mm); bypass valve `BV-1` on the canister lid.
- **Electrical:** 230 V, 50 Hz, 3.2 A; connected to the red (emergency power) sockets only.

3. TIME TO REACH THE TARGET OXYGEN LEVEL

When the door is closed and oxygen flows in, the oxygen fraction inside the kennel rises towards the inlet value along a wash-in curve. The time needed to reach a target fraction is estimated with equation (1).

$$t = \frac{V}{Q} \times ln\left(\frac{F_{in} - F_0}{F_{in} - F_T}\right) \qquad (1)$$

where t is the fill time (min), V the net gas volume of the kennel (395 L), Q the fresh-gas flow (L/min), F_{in} the oxygen fraction of the inlet gas (0.99 for wall oxygen), F_0 the starting fraction (0.21 for room air) and F_T the target fraction set on the controller.

Worked example. At Q = 15 L/min from room air to a target of 40 % O_2: V/Q = 395/15 = 26.3 min; ln((0.99 − 0.21)/(0.99 − 0.40)) = ln(0.78/0.59) = 0.279; t = 26.3 × 0.279 = 7.4 min. In fast-fill mode (30 L/min, allowed for the first 10 minutes only) the same kennel reaches 50 % O_2 in about 6.1 min.

4. CARBON DIOXIDE CONTROL

When the soda lime canister is in use, CO_2 is absorbed and the flow is chosen for oxygen only. When the canister is bypassed (valve `BV-1` on "OFF", for example during nebulisation or when the granules are exhausted), the flow itself must wash the CO_2 out. The CO_2 production of the patient is first estimated from body mass with equation (2).

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$$\dot{V}_{CO2} = k \times m^{0.75} \quad (2)$$

where $\dot{V}_{CO2}$ is the CO_2 production (L/min), k the clinic's resting coefficient of 0.011 L/min per $kg^{0.75}$ (use 0.016 for patients with a body temperature above 39.5 °C) and m the body mass in kg.

The minimum fresh-gas flow that keeps the CO_2 level in the kennel at or below the limit then follows from equation (3).

$$Q_{min} = \frac{\dot{V}_{CO2}}{x_{max} - x_{in}} \quad (3)$$

where Q_{min} is the minimum fresh-gas flow (L/min), x_{max} the permitted CO_2 fraction in the kennel (0.010, i.e. 1.0 %; until rev. B the limit was 0.5 %) and x_{in} the CO_2 fraction of the inlet gas (0 for wall oxygen, 0.0004 when medical air is mixed in).

Worked example. A 20 kg Border Collie on wall oxygen with the canister bypassed: $m^{0.75} = 20^{0.75} = 9.46$, so $\dot{V}_{CO2} = 0.011 \times 9.46 = 0.104$ L/min, and $Q_{min} = 0.104/(0.010 - 0) = 10.4$ L/min. Set the flow meter to the next whole litre above the result, here 11 L/min.

5. OXYGEN SUPPLY AND BACK-UP CYLINDERS

During a pipeline shut-down a back-up cylinder is connected through regulator `RG-200`. The available running time of a cylinder is estimated with equation (5).

$$t_{cyl} = \frac{V_w \times (P - P_r)}{Q} \quad (5)$$

where t_{cyl} is the running time (min), V_w the water capacity of the cylinder (10 L for the clinic's back-up cylinders), P the gauge pressure read on the regulator (bar), P_r the reserve pressure that must stay in the cylinder (20 bar) and Q the set flow (L/min).

Worked example. A full cylinder at 200 bar with a flow of 10 L/min gives $t_{cyl} = 10 \times (200 - 20)/10 = 180$ min. A cylinder below 50 bar is not connected to a kennel; it is returned to the gas store (room A0.03) with a red "EMPTY" tag.

6. OPERATING LIMITS FOR PATIENTS

- Standard target range: 40 % to 60 % O_2. A setpoint above 60 % is allowed for at most 24 h and needs written approval of the ICU clinician on duty in the patient record.
- Minimum fresh-gas flow at any time: 4 L/min, even when the soda lime canister is in use and the target has been reached.
- Chamber temperature: 18 °C to 24 °C for dogs; 20 °C to 26 °C for cats.

Brachycephalic patients. For brachycephalic breeds (for example French Bulldog, Pug, English Bulldog, Persian cat) the chamber temperature must not exceed 21 °C and the fresh-gas flow must be at least 10 L/min at all times, whether or not the CO_2 absorber is in use. If Q_{min} from equation (3) is higher, the higher value applies.

7. CHECKS, MAINTENANCE AND RESPONSIBILITIES

- **Before each patient (ICU nurse):** check the door seal, the soda lime colour and the oxygen alarm (set at 5 % below the setpoint).
- **Weekly, every Monday (ICU nurse on early shift):** two-point calibration of the oxygen analyser at 21 % (room air) and 100 % (wall oxygen). Until 1 July 2026 this calibration was done monthly. Results go in the kennel logbook.
- **Every six months (Clinical Engineering):** electrical safety and door-seal leak test.
- **Faults:** report to Clinical Engineering on extension 4417 (weekdays 07:30 to 17:00); outside these hours via the switchboard. A faulty kennel is labelled "OUT OF USE".

Owner of this data sheet: Femke Brinkhorst, clinical engineer, Clinical Engineering Section. Clinical content approved by Dr Joost Veltkamp, head of the small-animal ICU. Next review: September 2027.

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